Vehicle Fault Classification

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Abstract

Detection and classification of vehicle faults from audio is a challenging pattern recognition problem due to high noise levels, class imbalance, and limited availability of labeled data. In this project, we investigate the possibility of classifying common vehicle mechanical fault categories. These include categories such as brake noise, engine knock, belt squeal, and wheel bearing failure. Using real-world audio recordings collected through screen recording and personal vehicle operation, we evaluate multiple classification approaches, including a convolutional neural network (CNN) using spectrograms and a Transformer sequence model using time-ordered spectral features. To address data scarcity and class imbalance, we further explore conditional Variational Autoencoder (cVAE)–based synthetic spectrogram augmentation. Models are evaluated using group-based cross-validation to avoid data leakage across audio segments extracted from the same source recording. Experimental results show that adding large quantities of synthetic data can harm performance due to distribution mismatch. But can also improve generalization for both CNN and Transformer models. This study demonstrates both the promise and the limitations of generative augmentation for real-world acoustic fault classification.

1 Introduction

Mechanical faults in cars or anything with a combustion engine often have distinctive sounds before catastrophic failure occurs. Squealing belts, whining transmissions, knocking engines, and grinding wheel bearings can often be identified by mechanics using just sound. Automating this process has significant positive possible effects for preventive maintenance, safety, and cost reduction, especially as vehicles become more and more complex but stay similar in sound.

For my pattern recognition approach, vehicle audio classification presents several serious challenges. First, real-world recordings are noisy, having road noise, wind, talking, and background audio. Second, fault events may be sparse in sound rather than a continuous noise, making it extremely hard to segment. And third, labeled datasets for specific vehicle faults almost don't exist publicly, especially for rare conditions. These things make the problem well suited for a project using modern machine learning techniques combined with careful data preprocessing and evaluation.

In this project, we explore the classification of vehicle mechanical conditions using audio recordings collected directly by me. Unlike many prior studies that rely on curated datasets or simulated faults, this work focuses on noisy, real-world data from diagnosis, DIY fixes and live vehicle operation. We will evaluate both convolutional and Transformer neural network models trained on spectrogram representations of the audio. Then lastly, we test whether synthetic data generated by a conditional Variational Autoencoder (cVAE) can improve either models performance with limited data. I'm using deep learning on spectrograms rather than traditional audio features because vehicle fault sounds are non-stationary and recorded under noisy, volatile conditions. Convolutional and Transformer models allow time frequency patterns and temporal structure to be learned directly from the data. A conditional VAE is explored instead of simple signal-level transformations to better preserve class-specific characteristics when labeled data is limited.

2 Related Work

Audio-based fault detection has been studied across multiple domains, including machinery diagnostics, speech processing, and environmental sound classification. Traditional approaches often rely on handcrafted features such as Mel-frequency cepstral coefficients (MFCCs), spectral centroid, and energy-based descriptors, combined with classifiers such as support vector machines or k-nearest neighbors.

Prior research on audio classification and fault/anomaly detection generally falls into feature-based ML, spectrogram-based CNNs, attention/transformer models, and augmentation (including synthetic generation).

1. **Piczak (2015) – CNNs on time–frequency audio "images" (ESC-50).**
   Piczak showed that treating audio features as 2D inputs (e.g., spectrogram-like representations) enables CNNs to learn local time–frequency patterns effectively. This is **similar** to our spectrogram-CNN framing, but **different** in that it targets general environmental sounds rather than vehicle fault signatures.

2. **Hershey et al. (2016) – Large-scale CNN audio classification (AudioSet-style scale).**
   This work showed that CNNs can scale strongly for audio tagging when trained on massive datasets. It is similar in using CNNs for audio recognition, but different because our setting is smaller and more domain-specific (vehicle audio), where data scarcity and class imbalance are bigger issues.

3. **Koizumi et al. / DCASE 2020 Task 2 – anomalous sound detection (machine condition monitoring).**
   DCASE's anomalous sound detection line focuses on identifying "abnormal" machine sounds, often with limited or no labeled fault examples. This is **similar** in goal (detect mechanical anomalies via audio) but **different** because our project is a supervised multi-class classifier rather than primarily unsupervised anomaly detection.

4. **Gong et al. (2021) – Audio Spectrogram Transformer (AST).**

AST demonstrates that transformer attention applied directly to spectrogram patches can match or beat CNN-based baselines, especially for capturing longer-range dependencies. This is **similar** to our motivation for sequence-aware modeling, but **different** because our baseline is a CNN pipeline and our focus is automotive faults rather than broad audio benchmarks.

5. **Park et al. (2019) – SpecAugment (time/frequency masking on spectrogram features).** SpecAugment shows that simple perturbations directly in the time–frequency domain can improve robustness without generating entirely new samples. This is **similar** in addressing limited data, but **different** from our cVAE approach because it augments by *distorting existing samples* rather than synthesizing new ones.

3 Data

3.1 Data Collection

All audio data used in this project was collected by me. Recordings were obtained through a combination of direct screen recording of diagnostic videos and real-world vehicle videos taken by me. Doing this let me capture authentic mechanical sounds under realistic conditions, including background noise and varying recording quality.

The dataset consists of multiple classes corresponding to vehicle conditions, including:
Brake noise, Engine knock, Idle engine sound, Power steering squeal, etc.

Each class is stored in its own directory, containing raw .wav files of varying length. The dataset is limited in size and does not include fine-grained annotations, the class-level labels are good enough for supervised learning and comparative evaluation. Segmenting recordings into overlapping windows enables effective use of the data while group-based cross-validation ensures that evaluation remains realistic.

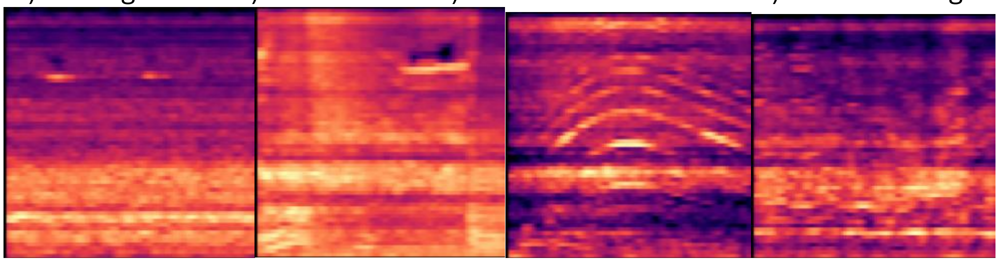
3.2 Preprocessing and Feature Extraction

Raw audio recordings are resampled to 16 kHz and then converted to mono. To handle variable length recordings and isolate informative parts, each audio file is segmented using a sliding window approach. Windows of 5 seconds are extracted with a stride of 1–2 seconds.

To reduce the influence of silence and background noise, an energy-based filter removes low-RMS segments and additionally for each window, the loudest subsegment is extracted to emphasize fault-related activity.

Each processed audio segment is converted into a spectrogram with 64 Mel bands. For the CNN model, log-scaled spectrograms are used directly. For the Transformer model, per-clip normalization is applied to stabilize training.

Example Spectrograms:
1.) Brake grind 2.)Turbo 3.)Transmission whine 4.) wheel bearing hum



4 Methods

4.1 Convolutional Neural Network

The CNN uses Mel spectrograms as 2D images with time and frequency axis. The network consists of multiple convolutional blocks with batch normalization and max pooling, followed by global average pooling and a fully connected classification layer. This design allows the model to learn local spectral patterns while remaining computationally efficient.

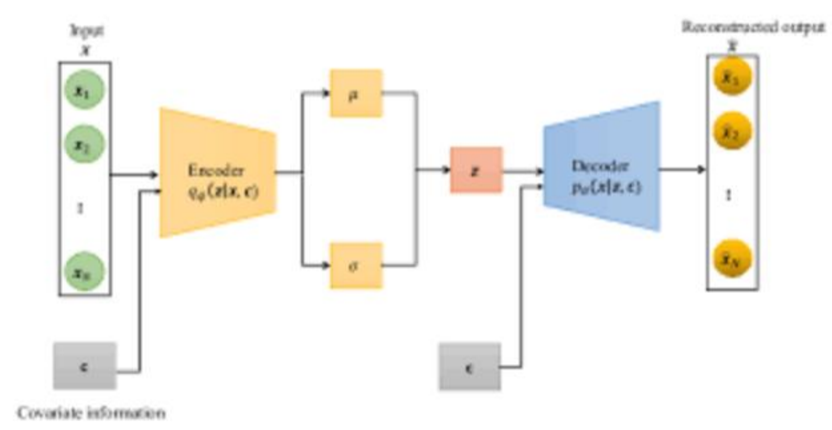
4.2 Transformer-Based Sequence Model

The Transformer model processes spectrograms as sequences of time frames, where each frame is represented by a Mel-frequency feature vector. A linear projection maps input features into a higher-dimensional embedding space, followed by positional encoding to preserve temporal order. Multiple Transformer encoder layers model long-range dependencies across time before pooled representations are passed to a classification head.

4.3 Synthetic Data Augmentation with cVAE

To address limited data and class imbalance, a conditional Variational Autoencoder is trained on real Mel spectrograms. The cVAE learns a latent representation conditioned on class labels, allowing it to generate new synthetic spectrograms for each fault category.

Experiments showed that naively adding large numbers of synthetic samples degraded model performance, likely due to distribution mismatch and over-smoothing in generated spectrograms.
(Illustration of cVAE logic)



5 Experiments and Results

5.1 Evaluation Protocol

Models are evaluated using group-based k-fold cross-validation, where all segments derived from a single source recording are assigned to the same fold. This prevents data leakage and promotes realistic generalization.

Accuracy is used as the primary evaluation metric. All experiments are conducted using identical preprocessing and evaluation settings to ensure fair comparison.

5.2 Baseline Results

Both CNN and Transformer models achieve above-random performance on the real-world dataset, with the Transformer generally outperforming the CNN due to its ability to model temporal dependencies.

```
Building dataset...
Total clips: 604
Class counts: Counter({0: 120, 2: 70, 3: 66, 8: 66, 6: 62, 9: 53, 4: 46, 7: 44, 1: 40, 5: 37})

Fold 1
Fold 1 acc: 0.3218

Fold 2
Fold 2 acc: 0.2289

Fold 3
Fold 3 acc: 0.1294

CNN GroupCV mean=0.2267 std=0.0786
```

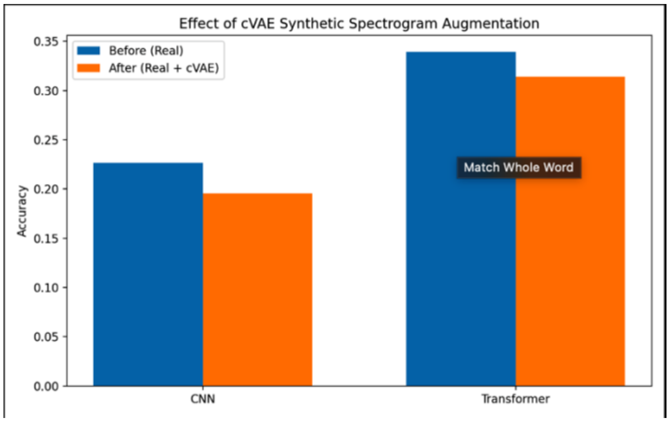
^ CNN BASLINE

```
/Users/paulclarkiv/PatternRec Final/venv/lib/python3.13/site-packages/torch/nn/modules/transformer.
  output = torch._nested_tensor_from_mask(
Building dataset...
Total clips: 604
Class counts: Counter({0: 120, 2: 70, 3: 66, 8: 66, 6: 62, 9: 53, 4: 46, 7: 44, 1: 40, 5: 37})

Fold 1
Fold 1 acc: 0.4752

Fold 2
Fold 2 acc: 0.2786

Fold 3
Fold 3 acc: 0.2637

Transformer GroupCV mean=0.3392 std=0.0964
```

^TRANSFORMER BASELINE

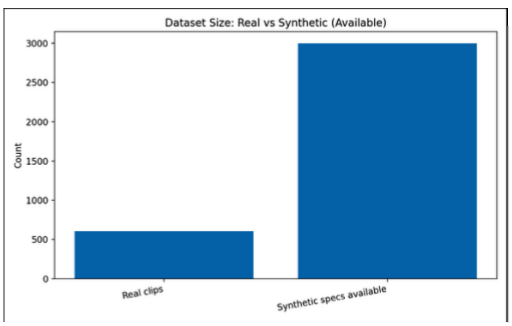
5.3 Effect of Synthetic Augmentation

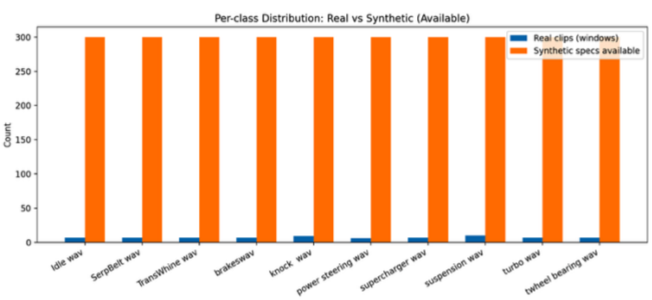
When large amounts of synthetic spectrograms are added, performance decreases for both models initially going from 30% to around 10%, confirming that low-quality generative augmentation can harm learning. But after restricting synthetic samples to a smaller proportion of the training data, performance improves relative to baseline.



5.4 Dataset Composition Analysis

To contextualize the results, we analyze the number of real and synthetic samples used during training, as well as per-class distributions.





6 Limitations and Ethical Implications

This work is limited extremely by data and recording diversity. All data was collected by just me, which can also introduce recording bias. Additionally, cVAE-generated spectrograms due to volatile data produce equally unreliable data as the real-world recordings. Adjusting ratio of synthetic spectrograms based on live, lager computation capability, larger dataset, and refined dataset, would all prevent the synthetic spectrograms from failing to add to the accuracy.

From an ethical standpoint, automated fault detection systems should not replace professional diagnostics anytime soon with such little data or without appropriate validation. Misclassification of vehicle faults could lead to unsafe decisions if relied upon without human oversight.

7 Conclusion and Future Work

This project shows that real-world vehicle fault classification from audio is actually possible using modern deep learning techniques, even under noisy and data-limited conditions(as long as you have the data). CNN and Transformer models both perform reasonably well, with Transformers showing stronger generalization. Synthetic data augmentation using cVAEs can be beneficial if carefully controlled, but naïve implementation will severely degrade performance.

Future work will explore higher-fidelity generative models, such as diffusion-based audio generation, incorporate multimodal inputs (e.g., vibration or OBD-II data), and expand the dataset to include a wider variety of vehicles and recording conditions.

8 Refrences

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